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THEMESYNC: A GRAPH-BASED DESIGN ORCHESTRATION FRAMEWORK WITH REAL-TIME CONSTRAINT SATISFACTION AND STATEFUL REFINEMENT

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By
Guiller Angelo Hermoso
Rona Mae Quite
Michelle Tercero

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CHAPTER 1

INTRODUCTION

This chapter will serve as an introduction to the research topic and highlights why it is relevant. This chapter provides the study's background by discussing and identifying the research problems and outlining the research objectives and questions. It also highlights the study's significance, scope, and limitations. Furthermore, this chapter includes a project dictionary defining key terms and concepts used throughout the study.

Background of the Problem

In the fast changing digital landscape, the creative industry is seeing an unprecedented increase in demand for automated design solutions. The market for AI-powered design tools is expected to increase from $5.54 billion in 2024 to $6.77 billion in 2025, representing a compound annual growth rate (CAGR) of more than 22% [16]. The growth is driven by the requirement for rapid asset development and cross-platform consistency, which aligns with UN Sustainable Development Goal 9, which stresses innovation, and Goal 12, which focuses on minimizing redundant design labor. Maintaining visual brand consistency, such as logos, color palettes, and typography, is no longer an aesthetic choice for businesses but a strategic imperative for establishing consumer trust and loyalty.

Historically, design tools have transitioned from manual, coordinate-based software like Adobe Photoshop to more accessible, template-driven platforms like Canva. This progression toward intelligent design is further exemplified by emerging industry research such as Pomelli by Google Labs. Pomelli utilizes semantic modeling to map high-level aesthetic descriptions directly to component styles, allowing for a more intuitive translation of creative intent [6]. However, a significant "Generative Gap" remains. Current industrial tools

often treat designs as stochastic or static entities. When a design is exported, it is typically ”flattened” into pixel-based formats (JPEG/PNG), which effectively destroys the underlying ”Visual DNA”—the mathematical and relational logic of the elements. This leads to Destructive State Management, where any manual refinement made by a user is lost the moment a global theme update is applied.

Furthermore, existing tools frequently suffer from Layout Fragility across Heterogeneous Dimensions. Many industrial tools rely on absolute coordinate-based layouts that lack relational logic. When a user transitions from a vertical format (like an ID) to a horizontal one (like a certificate), the system often fails to determine optimal placement, leading to Responsive Layout Failures (RLFs). These failures include elements protruding off edges or overlapping as layout space becomes scarce, making the output look unprofessional or non-functional [1, 21].

Also, while modern Generative AI models can produce high-fidelity imagery, they are often stochastic and computationally ”opaque”. These models can ”hallucinate” artistic elements but lack the Deterministic Vector Structures (SVG) necessary for precise, rule-based orchestration. There is a critical need for a system that treats a design not as a flat file, but as a Stateful Collection of Objects that can maintain visual harmony through mathematical constraints.

Within the context of academic and organizational environments, the need for consistent branding across different common publication materials is a recurring challenge. Without a framework that utilizes Graph Neural Networks (GNN) to understand structural hierarchies and Constraint Satisfaction Problem (CSP) Solvers to maintain layout requirements, producing these assets remains a labor-intensive process prone to human error. Without a shared relational model that encodes an organization’s stylistic and structural constraints, even minor brand updates — such as a color refinement or a logo revision — cascade into a full manual redesign cycle across every affected publication material. This gap highlights the pressing need for an automated, constraint-aware approach to brand consistency.

ThemeSync addresses these gaps by proposing a graph-based orchestration framework. By utilizing K-Means Clustering for color extraction and SIFT/SURF for feature matching, the system can autonomously identify a brand's stylistic markers. By integrating a Bi-directional State Machine, the framework ensures that manual user "nudges" are preserved as "Hard Constraints," allowing for a seamless "Human-in-the-Loop" experience that balances automated efficiency with creative control. By treating every design asset as a node within a relational graph, brand updates propagate across publication formats.

Statement of the Problem

The rapid transition to multi-platform digital branding demands tools that can seamlessly maintain visual integrity across heterogeneous formats. However, current automated design systems rely heavily on static templates or stochastic generation, creating a significant "Generative Gap" where outputs are rendered as computationally opaque, flat files (e.g., JPEG or PNG). This limitation leads to destructive state management, meaning any manual refinement or "nudge" made by a user is mathematically irreconcilable with the master theme and is subsequently lost during global updates. Furthermore, because these industrial tools utilize absolute coordinate-based layouts rather than relational logic, they suffer from layout fragility; transitioning a design between vastly different aspect ratios frequently results in broken layouts that require extensive, labor-intensive manual correction.

To address these computational bottlenecks, this study proposes ThemeSync, a specialized design orchestration framework that transitions away from immutable files by treating layouts as a stateful collection of objects. The proposed method utilizes a Constraint Satisfaction Problem (CSP) Solver, specifically the Cassowary Algorithm, to treat manual user edits as "Hard Constraints," triggering real-time recalculations to maintain visual harmony among surrounding elements. Coupled with Graph Neural Networks (GNN) to map struc-

tural dependencies and a Bi-directional State Machine to serialize data, the system ensures that local human-in-the-loop refinements and global theme updates are mathematically synchronized without data loss.

Ultimately, the purpose of this research is to develop and evaluate this orchestration framework to bridge the gap between automated synthesis and manual refinement. By doing so, the study aims to answer how Graph-Based DNA Extraction can successfully preserve stylistic markers across varying formats, and to what extent a CSP solver can effectively manage real-time manual edits while maintaining global design rules.

Objectives of the Study

General Objective

This study aims to design and evaluate ThemeSync, a graph-based design orchestration framework that implements real-time constraint satisfaction and stateful manual refinements across heterogeneous design formats

Specific Objectives

To achieve the general objective, the researchers aim to accomplish the following:

1. To design a Relational Logic and Feature Extraction model integrating Graph Neural Networks (GNN), K-Means Clustering, and SIFT/SURF algorithms to autonomously identify and map structural dependencies and stylistic markers from master design files.

2. To develop a Reactive Layout Synthesis engine utilizing a Bi-directional State Machine and the Cassowary Algorithm to solve Constraint Satisfaction Problems (CSP), enabling the real-time propagation of manual ”Hard Constraints” while preserving global layout states.

3. To evaluate the architectural integrity and professional viability of the generated multi-format layouts by implementing a mathematically calibrated Heuristic Fitness Function to quantitatively rank and optimize the aesthetic validity of parallel design candidates, followed by a quantitative human-expert validation.

Significance of the Study

This study aims to enhance the traditional design workflow by integrating a graph-based orchestration framework, which therefore improves the efficiency of the layout, reduces manual labor, and maintains mathematical visual harmony among all the formats. This study will benefit the following:

Students and Faculty Designers. Designing school-wide materials such as IDs, certificates, and posters often becomes inefficient due to repetitive resizing and layout inconsistencies. A system for creating ”Master Themes” that automatically readjust themselves for the appropriate size reduces the need for manual formatting of each file, and allows students and faculty to concentrate on subject matter and creative aspects, rather than on avoiding technical errors.

Camarines Sur Polytechnic Colleges. The college will benefit from a more efficient process for producing school materials. Automating design tasks allows the institution to save time and effort on large projects while ensuring that official documents remain consistent, accurate, and professional. This reduces the operational burden on staff responsible for managing and producing institutional publications.

Creative Professionals and Industrial Organization. Current design tools and production workflows often require manual, repetitive adjustments when resizing assets for various platforms, leading to ”Layout Fragility” and high labor costs. This study will provide a specialized pipeline for Multi-Format Asset Orchestration and utilizes a Constraint Satisfaction Problem (CSP) Solver to automate the scaling of ”Visual DNA.”

Researchers. The study will contribute to the fields of Intelligent Systems and Design Automation by exploring the intersection of Graph Neural Networks (GNN) and Deterministic State Orchestration. It will provide a rigorous methodology for Object Serialization and Bi-directional State Machines, offering new academic insights into how design intent—traditionally a subjective human trait—can be mathematically preserved and synchronized within a digital framework.

Future Researchers. The system will establish a technical foundation that will allow future scholars to enhance the automation engine or adapt its algorithms for other digital media. Developers and scholars will be able to utilize the study's findings on Heuristic Search and Aesthetic Scoring to build more advanced, user-centric technologies or investigate the potential of real-time collaborative design environments.

Computing Field. The paper will improve the technology of Artificial Intelligence (AI) and Computer Vision by utilizing K-Means Clustering for color extraction and Heuristic Search Algorithms for real-time layout suggestions. It will demonstrate how a Human-in-the-Loop (HITL) approach can combine mathematical precision with human creativity. The proposed research will offer a viable solution to the "Generative Gap" by converting flat, "opaque" files into Stateful Vector Structures, thereby enhancing the capabilities of smart design environments.

Scope and Limitation

The primary scope of this research involves the design, development, and technical evaluation of ThemeSync, a graph-based design orchestration framework engineered to automate brand identity synchronization across heterogeneous formats while addressing the challenges of "Generative Gap" and "Layout Fragility" in conventional design tools. Specifically, the study seeks to design a Relational Logic and Feature Extraction model that integrates Graph Neural Networks, K-Means Clustering, and SIFT/SURF algorithms

to identify and map structural dependencies and stylistic markers from master design files. Furthermore, the scope encompasses the implementation of Dynamic Node Instantiation, enabling the framework to merge extracted ”Visual DNA” with external administrative datasets—like user-defined JSON payloads containing specific names, titles, or signature counts—to generate fully populated, variable-driven documents. It also aims to implement a Reactive Layout Synthesis engine powered by a Bi-directional State Machine and the Cassowary Algorithm to solve Constraint Satisfaction Problems, ensuring real-time propagation of manual “Hard Constraints” and preservation of global layout states during resizing or format transitions. Lastly, the research intends to evaluate the architectural integrity and aesthetic quality of generated multi-format layouts through heuristic search and a mathematically calibrated fitness function, thereby optimizing structural consistency and visual validity across diverse platforms.

The study is limited to the synchronization of 2D graphical layouts and branding assets; it does not extend to 3D modeling or video-based motion graphics. Furthermore, while the system will provide ”Aesthetic-Aware” suggestions, the final design authority will remain with the human user, focusing on a Human-in-the-Loop refinement process rather than autonomous design replacement. Consequently, while the core computational evaluation of ThemeSync will rely on rigorous mathematical benchmarking—specifically measuring algorithmic latency, spatial prediction accuracy via Intersection over Union (IoU), and constraint satisfaction rates—the validation of the final generated layouts will not rely exclusively on theoretical metrics. To ensure the framework’s outputs align with real-world professional standards, the study will integrate a mixed-methods evaluation strategy. The mathematically validated design candidates will be subsequently evaluated by a human domain expert (a professional layout artist) using a quantifiable metric. This approach guarantees that the framework meets both strict computational constraints and practical industry aesthetic requirements.

Project Dictionary

This section will provide the definitions of the major terms and concepts that will be used throughout this study, serving as a guide in understanding the technologies, methods, and components involved in developing ThemeSync.

Affine Transformations. An affine transformation is the group of transformations that leave the parallelism of straight lines fixed, it can be combined with any transformation consisting of rotation, translation, shearing, and scaling [17]. In this study, Affine Transformations are mathematical functions that are used to transform coordinates and hierarchies of visual images between the master design and new canvas sizes in heterogeneous aspect ratios.

Bi-Directional state machine. A state machine is a form of automation where the execution of the system operates with a finite amount of states and transitions to another state based on an activity or condition [19]. In this study, the calculation algorithm is the one that is used to coordinate the refinements made manually with the automated layouts by sequentializing the refinements into a stable JSON state.

Cassowary algorithm. Cassowary is an incremental constraint-solving system used to handle automated model generation and linear equalities within complex configuration problems [14]. In this study, this algorithm is the main layout engine that re-calculates soft constraints when it is injected with manual Hard Constraints to ensure that design elements are not overlapping.

Constraint Satisfaction Problem (CSP). Constraint Satisfaction Problem (CSP) are core in artificial intelligence and theoretical computer science, a model of variables with domains in finite sets and a list of constraints, which are to be satisfied [5]. In this study, the mathematical model used to control layout density and support recalculation of design nodes in real time to create visual harmony is CSP. This enables the system to dynamically enforce branding constraints across all generated publication materials.

Dynamic Node Instantiation. Dynamic Node Instantiation is the algorithmic process by which external, user-defined variables are assigned baseline geometric dimensions and converted into functional mathematical nodes. The JSON format acts as a critical standard for lightweight data exchange, and its nested structure is uniquely suited for representing these diverse UI properties [2]. In this study, Dynamic Node Instantiation enables seamless communication between the frontend interfaces and the backend Graph Neural Network, allowing dynamic data to be orchestrated alongside the master theme.

Fitness function. Fitness Function. This is a quantitative figure of merit used to evaluate how effectively a candidate solution satisfies established design composition features and structural requirements [7]. In this study, the mathematical equation to rank the layout candidates is the Fitness Function, which is used to determine how well the layouts meet design rules such as the rule of thirds and the ratio of whitespace to other elements.

Graph Neural Network (GNN). Graph Neural Network (GNN) is a type of neural network that represents graph dependencies by exchanging messages among graph nodes in order to address problems that involve non-Euclidean space [3]. In this study, the GNN is the machine learning model that is applied to analyze design hierarchies where elements are viewed as nodes and the relationship between elements as edges to determine how best to position elements.

Hard Constraint. Hard Constraint is a requirement that must be strictly satisfied in a constraint optimization problem [20]. In this study, any manual user edit, such as moving or resizing an element, is registered as a "Hard Constraint" that the system must respect while adjusting other elements.

Implicit Geometric Hierarchy. Implicit Geometric Hierarchy is the mathematical principle by which the framework establishes visual priority. Graph Neural Networks are capable of capturing irregular object boundaries and complex spatial relationships with high accuracy by treating distinct areas or patches as individual graph nodes. [8, 10] this study, rather than relying on explicit semantic tags, the GNN infers spatial dominance by

evaluating the mathematical area of a node's bounding box to achieve high-precision UI element alignment. Elements instantiated with larger geometric footprints are mathematically prioritized and positioned accordingly within the structural hierarchy.

K-Means clustering. A basic activity within data analysis is clustering, which sets out to arrange a sizeable dataset into significant subsets along the similarity measures, with the K-Means algorithm being modest in the sense that the algorithm ascertains the data points and assigns them to the closest cluster centroid [11]. In this study, K-Means Clustering is the algorithm that is utilized in order to automatically identify dominant and accent color palettes in the pixels of a Master Design file.

Mean Opinion Score (MOS). A standardized, quantitative measure used to evaluate the subjective quality or experience of a system's output by human experts [18]. In this study, MOS refers to the structured 5-point Likert scale rubric utilized by the layout expert to mathematically grade the aesthetic validity, visual hierarchy, and brand compliance of the algorithmically generated layouts.

Object Serialization. Object Serialization. Object Serialization is the process of transforming complex data structures into a standardized format to enable real-time synchronization and system architecture persistence [15]. In this study, Object Serialization refers to the transformation of visual design objects into a structured JSON format to preserve a "Live State Link" so that these objects can be post-generatively editable.

Semantic Layout Parsing. Semantic Layout Parsing is the process of identifying and categorizing different parts of a visual layout to understand its underlying structure [4]. In this study, it is a pipeline used for PNG inputs that utilizes object detection (YOLO) and OCR to "unflatten" raster images into editable semantic nodes.

SIFT/SURF (Scale-Invariant Feature Transform / Speeded Up Robust Features). SIFT/SURF are algorithms in computer vision used to detect and describe local features in images [12]. In this study, these tools are applied to establish stylistic features and DNA, line weights, and textures to maintain a visual consistency throughout the vector library.

Soft Constraints. Soft Constraints are preferences or goals within a layout system that are not strictly required for a valid solution but are optimized to achieve the most desirable aesthetic or functional outcome [11]. According to Jiang et al. (2021), these constraints allow for flexible user interface layouts that can adapt to varying screen dimensions while maintaining proportional harmony. In this study, Soft Constraints represent the initial automated placement of design elements which can be overriden or re-calculated when a user introduces manual Hard Constraints.

Visual DNA. An algorithm to encode images as distributions of neuron activations (DNAs) to form an expressive, high-dimensional blueprint enabling visual attributes in one dataset to be compared and preserved in another dataset [13]. In this study, Visual DNA will refer to a hybrid data structure combining autonomously extracted algorithmic features (color centroids and relational logic) with user-defined administrative constraints (specific font families, weights, and baseline sizes).

Notes

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CHAPTER 2

RELATED LITERATURE AND STUDIES

This chapter is separated into three parts: the related literature and studies, synthesis of the state-of-the-art, and the gap bridge by the study. It would be important in creating a base for the study, increasing the researchers' field knowledge, and analyzing the research in connection to existing literature and studies.

Graph-Based Relational Logic and Perceptual Extraction

The evolution of automated design frameworks has progressed from basic image matching toward high-precision perceptual analysis. While Shaharom and Tahar [2023] established the foundational reliability of SIFT and SURF algorithms for maintaining structural integrity and asset preservation during image matching, Garcia and Chen [2023] enhanced the precision of these systems by shifting toward the CIE L*a*b* color space for superior perceptual extraction. This transition facilitated a more human-centric approach to identifying brand signatures by separating luminance from chromaticity, thereby effectively reducing visual noise.

Building upon these extraction techniques, Lertrusdachakul, Meesad, and Kittirachakul [2025] addressed the mathematical limitations of earlier clustering models by proposing a Hybrid Generalized K-Means approach to stabilize color centroid determination. Finally, Huang, Chen, and Liu [2025] demonstrated the current versatility of these frameworks by applying intelligent palette mapping to automate complex, high-fidelity design transitions. This progression ensures that modern systems can perform stylistic changes while strictly adhering to the structural ”DNA” of the original brand identity.

Graph Neural Networks (GNN) in Spatial Prediction

The transition toward automated design systems began with the foundational challenge of representing implicit spatial logic through graph-structured data. While Lee, Jin, and Tseng [2023] introduced the Neural Design Network (NDN) to demonstrate how graph convolution kernels could mimic human spatial reasoning, Nauata, Chang, and Cheng [2020] established that relational generative adversarial networks (House-GAN) could encode complex spatial adjacency as bubble diagrams. These early works effectively shifted the field away from rigid, rule-based systems toward models capable of learning the fluid relationships between design elements.

Building on these relational foundations, Nauata, Hosseini, and Chang [2021] evolved this technology into House-GAN++, a refinement network that moved beyond simple adjacency to demonstrate how GNNs could iteratively transform relational graphs into structurally consistent architectural layouts. By 2024, the application of GNNs expanded into specialized modeling; Jiang, Stuerzlinger, and Richter [2020] introduced Graph4GUI, utilizing bipartite graphs to achieve high-precision UI element alignment. This was further validated by Purohit, Vyas, and Gupta [2024], who confirmed that GNN-based optimization achieves superior performance in constrained residential setups compared to traditional metaheuristic baselines.

Most recently, Kamboj, Kumar, and Deep [2025] culminated this research by demonstrating an end-to-end pipeline in Graphs to Blueprints, capable of converting abstract graph nodes into consistent floor plans at a massive scale. Graph-encoded relational constraints ensure architecturally coherent outputs at scale. This progression—from the initial mimicking of human reasoning to the achievement of high-accuracy UI prediction and large-scale automated modeling—represents the complete evolution of the spatial logic required for modern design orchestration, moving from abstract concepts to a high-precision, mathematically-driven reality.

The development of document image understanding reached a significant turning point with the introduction of LayoutLM by Xu, Li, Cui, Huang, Wei, and Zhou [2020]. Their research established the methodology of treating text blocks, layout positions, and visual embeddings as semantic nodes, defending the principle that raw page images can be decomposed into meaningful units like paragraphs, figures, and tables. This foundational approach allowed graph-based models to reason over both textual and visual structures simultaneously. By 2022, the field advanced as Han, Wang, Guo, Tang, Wu, and Tian [2022] generalized node extraction beyond documents in their work Vision GNN. By splitting raw images into non-overlapping patches that serve as graph nodes, they demonstrated that patch-based extraction is semantically richer than traditional CNN grids. This methodology enables Graph Neural Networks (GNNs) to capture irregular object boundaries and complex spatial relationships with higher accuracy. Building upon these foundations, Wang, Xu, Cui, and Wei [2023] extraction by explicitly modeling text, image, and structural components as heterogeneous nodes within a graph. This progression illustrates a clear trajectory: from the initial establishment of semantic node extraction in 2020, through the extension to general vision tasks in 2022, to the recent culmination in hybrid systems. These modern advancements enable GNNs to reason over complex document and image structures with significantly greater semantic fidelity.

Before a Graph Neural Network can reason over spatial layouts, the raw visual data must first be transformed into discrete, mathematical nodes. Recent literature demonstrates that combining high-speed object detection architectures with Optical Character Recognition (OCR) establishes a robust pipeline for Semantic Layout Parsing. A recent study evaluating the comparative performance of YOLOv8 for the layout analysis of document images concluded that the YOLOv8 architecture provides state-of-the-art precision and speed for extracting structural bounding boxes across complex, heterogeneous layouts [2025]. Building upon this validated extraction method, the proposed methodology will pipeline these generated bounding boxes into the Tesseract OCR engine to retrieve the corresponding

textual representations. This hybridized extraction methodology will effectively unflatten raster images, providing the precise spatial embeddings and textual nodes required for downstream GNN processing without relying on manual data entry.

Constraint Satisfaction Problems (CSP) in Design

The journey toward mathematically maintaining structural integrity during automated layout transformations began with addressing the limitations of traditional linear solvers. Jiang, Stuerzlinger, Zwicker, and Lutteroth [2020] developed the ORCSolver, an efficient system designed specifically for adaptive GUI layouts containing non-convex OR-constraints. This breakthrough allowed design frameworks to handle complex scenarios—such as selecting between disparate navigation styles based on viewport space—moving beyond static grids to establish the groundwork for responsive environments that preserve semantic relationships across platforms.

As the demand for more fluid user experiences grew, research shifted toward achieving sub-millisecond response times. Li, Li, Fu, and Cai [2024] introduced SMT-Layout, a MaxSMT-based approach engineered to support real-time interaction in complex interfaces. By formulating layout rules as Satisfiability Modulo Theories (SMT) problems, the system gained the ability to dynamically resolve both soft and hard constraints. This allowed design elements to resize instantly during user interaction, successfully bridging the gap between theoretical constraint solving and practical, high-speed GUI behavior.

The most recent advancements have focused on the dual pillars of scalability and automation. To address the computational demands of modern interfaces, Zhang, Maqrot, Mouysset, and Bortolaso [2025] introduced a CSP-based scalable framework that utilizes decomposition techniques to manage hundreds of interdependent variables without lag. Simultaneously, the manual effort required to define these relationships has been reduced through the work of Penko, Penko, Varga, and Šoštarić [2025], who developed a Large

Language Model-driven framework capable of translating natural language requirements into formal constraint models. Together, these developments represent a complete progression from solving specific adaptive logic to enabling the scalable, automated generation of complex, real-time design models.

Stateful Refinement and Iterative Systems

The development of sophisticated orchestration frameworks begins with the foundational selection of data formats capable of representing complex hierarchical structures. Berdan [2025] established the JSON format as a critical standard for lightweight data exchange, identifying its nested structure as uniquely suited for representing the diverse properties of UI nodes. This structural backbone enables seamless communication between frontend interfaces and backend Graph Neural Network (engines), setting the stage for more complex state management. Once this communication channel is established, the focus shifts to maintaining consistency during automated adjustments. Prymushko, Puchko, Yaroshynskyi, Sinko, Kravtsov, and [2025] addressed this by implementing Conflict-Free Replicated Data Types (CRDTs), which provide a mathematical guarantee for efficient state synchronization in distributed environments. By utilizing CRDTs, a design system can resolve conflicting layout changes without requiring a central coordinator, ensuring the design state remains consistent across all active environments.

As synchronization logic became more robust, research moved toward optimizing the speed of these transitions. Gomes, Rodrigues, and Baquero [2025] introduced the ConflictSync approach, which utilizes a mechanism for bandwidth-efficient synchronization of divergent states. By focusing on transmitting delta updates rather than full state snapshots, this methodology ensures that complex design graphs can be updated in real-time even across low-bandwidth connections. This efficiency gain proves critical in collaborative environments where multiple agents concurrently modify shared design states.

This technical efficiency is a prerequisite for the final stage of orchestration: the iterative improvement of the system itself. To evaluate this capability, Cheng, Tang, Ma, Hackett, He, Su, Beschastnikh, Huang, Ma, and Xu [2025] introduced SysMoBench, a comprehensive benchmark for evaluating AI performance when formally modeling complex real-world systems. Their research emphasizes that refinement is a series of stateful transitions that gradually eliminate design ”smells” or constraint violations. By integrating these iterative heuristics with high-speed synchronization, modern frameworks can ensure that every refinement cycle brings the layout closer to the target brand identity while maintaining the structural integrity of the original design intent.

The evolution of bi-directional state machines in user interface research can be traced from early cyber-physical synchronization studies to subsequent applications in human-facing systems. Liang, McGee, Menassa, and Kamat [2020]established the principle that digital twins and physical robots must maintain continuous reciprocal updates, effectively modeling the interaction as a bi-directional finite state machine. This work highlighted the fundamental necessity of two-way state propagation to ensure total fidelity between simulated models and physical environments.

Building on this foundation, Goldenberg and Tractinsky [2021] extended the theoretical defense of bi-directionality into the domain of graphical user interfaces. Their study demonstrated that multilingual interfaces—particularly those requiring transitions between right-to-left (RTL) and left-to-right (LTR) orientations—inherently demand two-way directional state transitions. They argued that such interfaces should be formally understood as bi-directional state machines where layout and text orientation dynamically toggle in response to user context. Taken together, these studies illustrate a clear progression: while the foundational work in 2020 defended bi-directional synchronization in cyber-physical systems, subsequent research in 2021 applied the same theoretical framework to multilingual UI design. Subsequent work further extended this framework beyond UI design, exploring how bi-directional state machines could support adaptive accessibility systems.

This trajectory grounds bi-directional state machines as a unifying principle that ensures consistency across both machine-to-machine and human-centered interfaces.

Orchestration and Real-Time Processing

The successful orchestration of automated layout generation within a live, interactive canvas necessitates strict computational performance boundaries. Du, Turner, Dzitsiuk, and Kowdle [2020] established that complex geometric operations must be processed within strict millisecond thresholds to prevent user interface occlusion or visual freezing, a principle critical for the viability of algorithmic design frameworks in real-world applications. Furthermore, Hu, Wang, and Wu [2025] identified that as generative artificial intelligence paradigms scale to support highly complex spatial layout designs, the computational trade-off between generating optimal configurations and maintaining real-time responsiveness becomes a primary architectural constraint. Consequently, the system must be engineered to resolve mathematical graph dependencies at interactive rates to ensure the orchestration engine does not bottleneck the graphical interface.

To bridge the gap between static algorithmic prediction and dynamic manual refinement, the architecture implements a Human-in-the-Loop (HITL) orchestration paradigm. While traditional end-to-end models often execute procedural planning in a computational vacuum, He, Huang, and Wang [2022] mathematically proved that computational architectures can co-evolve spatial structures alongside human experts through their iPLAN model. By accommodating free-form interactions such as localized element nudging or resizing, the system treats manual adjustments as definitive, hardened constraints on the canvas. Instead of passively generating a new coordinate array that erases human input, the algorithm proposes procedural suggestions and calculates spatial harmonizations around the user's localized edits. This ensures the orchestration engine remains a collaborative instrument rather than an autonomous override mechanism.

By synthesizing these high-speed constraints and HITL principles, the orchestration layer will operate as a highly synchronized, event-driven ecosystem. To prevent computational bottlenecks and ensure the graphical interface does not suffer from visual occlusion or frame-rate degradation during manual manipulation, the framework will decouple continuous drag states from the mathematical solvers. Every manual human refinement will act as an asynchronous, post-interaction event trigger. Specifically, the system will only harden the spatial parameters and force the underlying constraint solver to calculate optimal coordinate placements upon the user's release of the element (the on-drop or on-resize-complete event hook). This event-driven approach will mathematically guarantee that the algorithm respects the interactive human inputs without overloading the CPU during fluid drag states, successfully maintaining both global layout harmony and strict latency thresholds [2025, 2020, 2022].

Algorithmic Evaluation and Aesthetic Scoring

The evaluation of automated layout architectures necessitates a robust synthesis of high-speed search algorithms and mathematically grounded aesthetic evaluators. Chen, Wang, and Tan [2021] identified that navigating the nearly infinite permutation space of graphical element placements poses a significant computational bottleneck, as standard linear search models and evolutionary algorithms frequently suffer from latency and stagnation. To resolve these inefficiencies, Vinod and Murthy [2022] utilized nature-inspired metaheuristic algorithms to drastically reduce the time-space complexity required for solving engineering layout problems. Specifically, Got, Moussaoui, and Zouache [2021] deployed hybrid filter-wrapper mathematical models, such as the Whale Optimization Algorithm, to treat feature selection as a multi-objective task, yielding superior convergence rates. Furthermore, Qais, Mirjalili, and Hasanien [2022] introduced geometry-based frameworks like the Circle Search Algorithm, which uses circular motion to optimize search trajectories.

Following the generation of spatial coordinates, the framework applies quantitative aesthetic evaluation to score the visual quality of the layouts. Zheng, Yang, and Zeng [2020] employed combined CNN-GNN architectures to extract local and global structural features, resulting in an objective compositional scoring system that aligns with strict mathematical rules. Similarly, Guo, Zhao, and Zhang [2025] demonstrated the use of Graph Neural Networks to quantify human-centric design rules—such as whitespace and visual density—into objective metrics. These intelligent design systems utilize AI-driven aesthetic scores to guide evolutionary strategies, successfully automating layouts that balance functionality with visual harmony [39].

Beyond structural alignment and whitespace ratios, modern automated design frameworks must also account for visual interference between generated elements and underlying background content. To prevent the degradation of text legibility caused by visual noise, the proposed layout optimization pipeline will integrate dynamic saliency evaluations. Drawing upon recent advancements in GAN-based domain adaptation for image-aware layout generation from the study of Xu, Chenchen and Zhou, Min and Ge, Tiezheng and Xu, Weiwei [2026], the framework will algorithmically quantify layout occlusion by computing the perceptual difference in the background's saliency map before and after the application of text masks. By establishing a rigid spatial clash metric (R_{shm}), the system will mathematically evaluate the aesthetic score of the composition. This process will ensure that the dynamically generated layout actively preserves the primary focal regions of the background while maintaining optimal structural legibility, effectively bypassing the need for static, offline feature extraction.

To manage the output of real-time generation cycles, the system must prevent the presentation of stochastic or biased candidates. Li, Chen, and Wang [2023] implemented fairness-aware recommendation ranking algorithms to mathematically balance and sort generated outputs based on rigorous merit. Ultimately, Shi, Shang, and Qi [2023]emphasized the integration of deep generative models to bridge the gap between high-fidelity co-

ordinate predictions and human-centric aesthetic preferences. This Aesthetic-Aware Preference Alignment (AAPA) provides the necessary heuristic filtering to validate optimal design structures before they are presented to the user.

Synthesis of the State-of-the-Art

Current state-of-the-art in automated layout generation and graphic design orchestration demonstrates a definitive paradigm shift from manual, coordinate-based drafting toward intelligent, algorithmically-driven frameworks [18, 30, 33]. Both the established literature and the proposed ThemeSync framework share a foundational reliance on computational algorithms to synthesize spatial arrangements. However, a critical synthesis of the reviewed literature reveals a persistent dichotomy between offline, generative Al models and real-time, interactive design systems. Deep generative models and graph-based frameworks excel at mathematically predicting spatial logic but predominantly operate as static, offline batch processors [23, 24]. ThemeSync departs from this passive generation model by embedding advanced computational algorithms into a live, stateful orchestration loop, transforming the canvas from a one-time predictive output into a continuously reactive environment.

This divergence is further evident in the treatment of visual identity and structural logic. Existing literature heavily emphasizes perceptually-conscious clustering models and feature matching algorithms—such as hybrid K-Means, SIFT, and SURF—to isolate dominant hues and stylistic markers with high precision [6, 31, 19]. Concurrently, Graph Neural Networks (GNNs) represent the current apex of relational spatial modeling, successfully predicting bounding boxes in offline environments [15, 41]. Yet, current frameworks frequently treat these domains in isolation; perceptual extraction is applied primarily to static image analysis, while structural GNNs independently predict layouts. ThemeSync synthesizes these disparate technologies to construct a unified, hybrid ”Visual DNA.” Current

automated design frameworks frequently suffer from typographic hallucinations, as relying purely on Optical Character Recognition (OCR) or feature matching to autonomously extract exact font families from flattened images is computationally error-prone. ThemeSync bridges this gap by decoupling structural extraction from typographic assignment. While the system integrates perceptual extraction directly with relational graph inference to autonomously map colors and stylistic identities onto a live structural graph, the exact typographic parameters—such as font families, specific font weights, and baseline geometric sizing—are deterministically defined by the user prior to orchestration. This ensures that institutional branding remains mathematically consistent during automated transformations, utilizing techniques akin to smart palette mapping [14] to move beyond flat visual adjustments and achieve deep structural awareness.

The most significant technological limitation in existing automated design tools is the prevalence of ”destructive updating,” where generative models force a complete layout recalculation that irreversibly overwrites manual user refinements. While recent computer science literature validates the necessity of Constraint Satisfaction Problems (CSP) for scalable spatial issues [20, 26, 39] and state machines for decentralized synchronization [22, 36, 25], their combined application to protect localized creative intent is notably absent in the design domain. ThemeSync bridges this gap by combining the incremental Cassowary constraint-solving algorithm [16] with a Bi-directional State Machine utilizing JSON-based object serialization [1]. Unlike traditional AI generators that act as ”fire-and-forget” black boxes, ThemeSync treats user interactions as strict ”Hard Constraints.” The serialized state machine ensures that these human-in-the-loop refinements survive global theme updates, utilizing the CSP solver to propagate non-destructive recalculations of the surrounding layout and eliminating layout fragility.

Finally, while the literature demonstrates that geometry-based metaheuristics [9, 34, 29] and CNN-GNN architectures [10, 40] can mathematically optimize and score layouts based on strict composition rules, finding an optimal layout programmatically does

not guarantee subjective user satisfaction. Contemporary generative models frequently force a single, mathematically prioritized output onto the designer. ThemeSync fundamentally contrasts with this approach by synthesizing high-speed state-space exploration with Aesthetic-Aware Preference Alignment (AAPA) [32]. Rather than automating the designer out of the workflow, the framework utilizes these heuristic-quality filters and fitness functions to evaluate and rank parallel layout candidates in real-time [21]. This interactive suggestion mechanism maintains the critical ”human-in-the-loop” paradigm, proving that high-speed algorithmic orchestration and stateful human creative control can be seamlessly integrated into a single design ecosystem.

Gap Bridged by the Study

Existing research in graphic design automation has largely predominantly advanced along distinct trajectories, with studies either focusing on automated layout generation or supporting manual designer adjustments, but rarely integrating both within a unified framework. As shown in a comprehensive survey on deep generative models [32], deep generative models are currently the most popular way to create intelligent layouts. However, these systems work as static, offline pipelines that act like generative black boxes, producing outputs that are structurally correct but cannot be changed by humans during the process. Any changes made by a user are automatically undone when the process is run again. This rigidity is further compounded by the absence of stateful memory in existing design tools, wherein no current system retains a record of designer intent, causing global operations such as theme or template updates to consistently and irreversibly erase localized refinements. Furthermore, even state-of-the-art diffusion-based layout models reduce generation to a purely predictive task, prioritizing mathematical constraints over the subjective aesthetic preferences of designers and failing to align outputs with the contextual demands of professional design practice [38, 25]. Taken together, these issues reveal a fun-

damental and unresolved gap between the operational efficiency of algorithmic automation and its capacity to support meaningful human creative control.

This study fills these gaps by introducing ThemeSync, a stateful graphic design orchestration framework that combines automated layout intelligence with non-destructive human refinement. In response to the offline processing limits identified in prior work [32], ThemeSync embeds multi-relational GNN inference and the Cassowary constraint-solving algorithm within a live orchestration loop, allowing structural changes to occur in real time while keeping relational dependencies intact. To resolve the destructive updating problem, the framework employs JSON-based state serialization and state machine logic to encode user interactions as hardened constraints, ensuring localized designer refinements persist through global synchronization events. Furthermore, contemporary deep generative layout models frequently suffer from semantic ambiguity when merging static structural templates with variable data. ThemeSync bridges this gap by decoupling the aesthetic extraction pipeline from a dynamic payload injection system. By utilizing Dynamic Node Instantiation and Implicit Geometric Hierarchy, the framework ensures that user-defined administrative variables—such as specific names or titles—are mathematically prioritized based on their assigned geometric baseline tensors, allowing dynamic content to be logically orchestrated alongside the master theme without relying on rigid semantic labels. Building on aesthetic alignment principles established in recent literature [25], ThemeSync integrates Aesthetic-Aware Preference Alignment (AAPA) and GNN compositional scoring into an interactive recommendation dashboard, replacing single programmatic outputs with a carefully selected set of aesthetically validated layout candidates. ThemeSync changes the role of AI in graphic design from an independent generator to a human-in-the-loop collaborative tool that respects both structural logic and creative intent.

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CHAPTER 3

METHODOLOGY

This chapter will discuss the research methods, procedures, and techniques to be used to systematically identify, select, process, and analyze information to address the specific problems posed in this study. It will explain theorems, algorithms, and mathematical models employed, the materials and statistical tools to be used, the theoretical/conceptual framework, and the methods for analyzing data to ensure accurate interpretation of results.

Research Design

This study will employ a Constructive Research Design, a methodological framework specifically utilized in Computer Science and software engineering to solve domain-specific problems through the creation of novel computational artifacts, algorithms, or mathematical architectures. Unlike descriptive or exploratory research that primarily observes existing phenomena, the constructive approach focuses on the systematic development and rigorous validation of a proposed solution. This design dictates that the research must yield a tangible, functional construct—in this case, an algorithmic framework—that addresses a defined technical gap, followed by an objective, mathematically grounded evaluation to prove the artifact's practical utility and theoretical soundness within the discipline.

In application to this study, observing or evaluating current design systems will be insufficient to solve the ”Generative Gap” and ”Layout Fragility” identified in contemporary graphic design automation. Because existing industrial tools rely on static templates that cause destructive state management, solving this problem requires the active construction of the ThemeSync orchestration architecture. By utilizing a constructive methodology, the researchers will systematically engineer the tri-layer pipeline, integrating Graph Neu-

ral Networks (GNN) for spatial mapping with a Cassowary linear arithmetic solver and a Bi-directional State Machine for non-destructive, real-time recalculations. Furthermore, to bridge the gap between theoretical computation and practical application, the design will integrate a mixed-methods evaluation paradigm. The framework's validity will be established by first mathematically proving the artifact's structural integrity against a Heuristic Fitness Function, followed by a quantitative human-expert validation using a Mean Opinion Score (MOS) to guarantee professional aesthetic compliance.

Theorems, Algorithms, and Mathematical Framework

This section will delineate the foundational theorems, computational algorithms, and mathematical models that drive the ThemeSync orchestration framework. Specifically, it will formalize the underlying architecture of the Graph Neural Networks (GNN) utilized for probabilistically mapping spatial dependencies, alongside the Convolutional Neural Networks (CNN) deployed for quantitative aesthetic evaluation. Furthermore, this section will explicitly define the system of linear inequalities and objective functions governing the Cassowary constraint-solving algorithm, establishing the mathematical boundaries required to dynamically calculate optimal layout coordinates while strictly preserving user-defined geometric parameters.

Perceptually-Consistent Color Quantization (K-Means Clustering)

ThemeSync uses K-Means Clustering in the CIE color space L*a*b, which is a perceptually uniform color space, unlike the non-linear RGB color space which is device-dependent, with the Euclidean distance between two points in the space directly proportional to the perceived difference between them by human vision. The algorithm divides the number of pixels in a picture, n, into k color clusters by minimizing the **Within-Cluster Sum of Squares (WCSS):** as shown in the formula below:

$$J = \sum_{i=1}^{k} \sum_{x_p \epsilon S_i} ||x_p - \mu_i||^2 \tag{3.1}$$

The extraction process defines x as the $L^*a^*b^*$ feature vector of an individual pixel, which is then grouped into clusters where i represents the mean of pixels within a specific cluster S_i [8]. These clusters ultimately determine k, the target color centroids representing the primary, secondary, and accent colors. By grounding the analysis in the $L^*a^*b^*$ color space, the system ensures that the resulting palette remains consistent across diverse display technologies; this is achieved because the Euclidean distance calculated between clusters through Formula 1 correlates directly with the visual difference perceived by the human eye [3].

Stylistic Marker Identification (SIFT/SURF)

To ensure the conservation of specific institutional resources, such as the CSPC logo, the system employs Scale-Invariant Feature Transform (SIFT) and Speeded-Up Robust Features (SURF). SIFT extracts local keypoints that are resistant to scaling, rotation, and affine transformation [15]. Mathematically, these keypoints are identified through a Difference of Gaussians (DoG) function as shown in Formula 2:

$$\mathbf{D}(x, y, \sigma) = (G(x, y, k\sigma) - G(x, y, \sigma)) * \mathbf{I}(x, y) \tag{3.2}$$

The application of Formula 2 allows the framework to recognize a logo in a high-resolution Master Design and accurately track its relative significance when transformed into heterogeneous formats [15]. Furthermore, the framework utilizes SIFT/SURF to construct a mathematical Saliency Map of the inpainted background layer. By aggregating the high-frequency keypoints detected via the DoG function—such as sharp gradients, dense textures, or disruptive visual noise—the system generates a spatial heat map. The orchestration engine subsequently utilizes this Saliency Map to calculate a Legibility Penalty,

ensuring that dynamically injected text nodes are mathematically restricted from occupying visually noisy coordinate spaces.

Linear Arithmetic Constraint Solving (The Cassowary Algorithm)

The Cassowary Algorithm serves as the core engine for real-time layout optimization. It is optimized to solve systems of linear equalities and inequalities efficiently [5]. Every design element is represented by a set of variables: $x_i\,(left)$, $y_i\,(top)$, $w_i\,(width)$, $h_i\,(height)$. Relational logic (e.g., margins and padding) is expressed through linear equations defined in Formula 3:

$$\begin{aligned} x_{\text{logo}} &= x_{\text{boundary}} + 20 \\ x_{\text{right}} &= x_{\text{left}} + \text{width} \end{aligned} \tag{3.3}$$

The solver uses a dual-simplex method to maintain these relationships established in Formula 3 incrementally, allowing the layout to re-calculate in milliseconds whenever a single variable—such as viewport width—is adjusted [9].

Stateful Human-in-the-Loop Refinement

The framework differentiates between two types of constraints based on mathematical priority to support real-time interaction [9]:

1. **Hard Constraints:** Manual user interactions, such as ”nudging” a text box or locking a logo’s position, are assigned infinite weight. These are strictly prohibited from being violated by the solver and are mathematically represented as a Hard Equality Constraint in Formula 4:

$$C_{\text{hard}} : f(x) = b \tag{3.4}$$

2. **Soft Constraints:** Standard design rules (e.g., default margins) or suggested alignments are assigned finite weights (w_i). The solver attempts to satisfy these by minimizing the total error through a Weighted Objective Function defined in Formula 5:

$$\textbf{minimize} \sum w_i d_i \tag{3.5}$$

Where d_i represents the "slack" or distance from the target suggested state [13].

Graph Neural Networks (GNN)

The architecture of a Graph Neural Network (GNN) will be designed for processing relational data by maintaining and updating node states through a graph structure [16]. In the ThemeSync framework, the design canvas will be modeled as a graph structure. Each design element—such as a text block, logo, or image—will function as a node, while spatial and semantic relationships—such as alignment or proximity—will be represented as edges. Through a message-passing mechanism, nodes will exchange information with their neighbors, allowing the system to capture dependencies among elements. This process will enable the model to infer how design components will interact and depend on one another, supporting relational logic inference for layout optimization.

Initial Feature Encoding

Every design element will start with a set of basic information called a feature vector [17]. This initial description will record the element's spatial and visual properties, serving as the starting point for relational analysis as established in Formula 6:

$$h_i^{(0)} = [x_i,\ y_i, w_i, h_i, c_i, s_i] \tag{3.6}$$

In this formulation provided by the formula above, (x_i, y_i) represents the specific position of the element on the canvas, while (w_i, h_i) denotes its corresponding width and

height dimensions. The term c_i refers to the CIE $L^*a^*b^*$ color embedding extracted via K-Means clustering, and s_i serves as the SIFT/SURF stylistic descriptor capturing the element's unique visual features. Because the baseline training dataset (CGL-Dataset) categorizes text under a generic supercategory rather than explicit semantic labels, the framework establishes visual priority through Implicit Geometric Hierarchy. During Dynamic Node Instantiation, the system algorithmically assigns Baseline Geometric Tensors (predefined bounding box dimensions, $w_i \times h_i$) to external administrative variables before they are encoded into the graph. The Graph Neural Network infers spatial dominance purely by evaluating these initial geometric parameters. Elements instantiated with a larger spatial footprint are mathematically assigned a higher relational weight by the GNN, ensuring they are positioned precisely at the top of the structural hierarchy without requiring explicit semantic tagging. By organizing these variables into the structure of Formula 6, the system creates a standardized input for the subsequent Graph Neural Network layers.

Relational Inference via Message Passing

The GNN will process these features through a layer-wise exchange [1]. At each layer k, every element will compute messages from its neighbors to update its own representation as defined in Formula 7:

$$\begin{aligned} m_{ij}^{(k)} &= \phi(h_i^{(k)},\ h_j^{(k)},\ e_{ij}) \\ h_i^{(k+1)} &= \psi(h_i^{(k)}, AGGm_{ij}^{(k)} :\ j\ \epsilon\ N(i)) \end{aligned} \tag{3.7}$$

In this architecture described by Formula 7, ϕ represents the message function, implemented as a multilayer perceptron (MLP) designed to compute pairwise relational features between adjacent nodes. The AGG term denotes a permutation-invariant aggregation operator, such as mean or sum pooling, while ψ serves as the node update function, utilized as a Gated Recurrent Unit (GRU) to update the hidden state of each node.

The connections e_{ij} utilized within the message function of Formula 7 will carry edge features describing spatial logic, which are formally structured in Formula 8:

$$e_{ij} = [\delta x_{ij}, \delta y_{ij}, IoU_{ij}, r_{ij}^{align}] \quad (3.8)$$

According to Formula 8, these features include distance, Intersection over Union (IoU), and an alignment score where 0 will mean no alignment and 1 will mean perfect alignment.

State Synchronization — Finite State Machine and JSON Schema

The State Synchronization module will record every change the designer will make during a session. Each interaction — whether a position nudge, a resize, or an element lock — will be saved as an immutable entry in a JSON-based state chain No prior state will ever be overwritten, which means the designer's complete editing history will be preserved and any earlier version of the canvas will be restorable. Each designer action will be modeled as a state transition. When an action will be performed at time t, the system will move from the current canvas state St to a new state St+1 based on a deterministic transition function defined in Formula 9

The State Synchronization module will record every definitive change the designer will make during a session. Each interaction—whether a finalized position nudge, a completed resize, or an element lock—will be saved as an immutable entry in a JSON-based state chain [14]. To ensure optimal computational latency, the state machine will operate strictly as an Event-Driven Post-Interaction Solver. The system will deliberately ignore continuous manipulation data (e.g., on-drag coordinate streams) to prevent overloading the solver pipeline. Instead, a state transition will only initiate upon a terminal user action. When a discrete event will be performed at time t, the system will move from the current canvas state S_t to a new state S_{t+1} based on a deterministic transition function defined in Formula 9 [6].

$$S_{t+1} = \delta(S_t, e_t) \quad (3.9)$$

In the model represented by Formula 9, S_t represents the current canvas state at time t, while e_t denotes the specific, terminal interaction event performed by the designer (re-

stricted exclusively to on-drop, on-resize-complete, or boolean lock toggles). The transition function is defined by δ, which processes the current state and the discrete event to trigger the Cassowary recalculation and produce the subsequent stabilized state of the canvas.

After each transition, every affected element will be assigned a constraint type [10]. The classification logic for these elements is formally presented in Formula 10:

$$\begin{aligned} c_i &= Hard,\ if\ \ e_i\ will\ belong\ to\ U_{manual} \\ c_i &= Soft,\ if\ \ e_i\ will\ belong\ to\ A_{algorithmic} \end{aligned} \tag{3.10}$$

As specified in Formula 10, elements manually adjusted by the designer (U_{manual}) will be labeled as Hard constraints, meaning the system will not change their position or dimensions. Conversely, elements placed or adjusted by the algorithm ($A_{algorithmic}$) will be labeled as Soft constraints, which the system will be permitted to modify during layout optimization.

Validation of Hybrid Visual DNA Correctness

To guarantee that the orchestration engine executes with absolute structural and stylistic precision, the framework will implement a dual-validation protocol to mathematically and programmatically ensure the correctness of the Hybrid Visual DNA prior to layout synthesis. Because the Visual DNA is composed of both extracted and injected variables, the validation will be bifurcated into Algorithmic Correctness and Administrative Correctness.

The correctness of the autonomously extracted color features will be mathematically validated by optimizing the Within-Cluster Sum of Squares (WCSS) during the K-Means execution, as established in Formula 3.1. To prevent the extraction of inaccurate or perceptual outlier colors, the algorithm will iteratively compute the WCSS across multiple centroid (k) configurations. The system will programmatically identify the optimal k-value by detecting the inflection point (Elbow Method) in the WCSS decay curve. This mathematical convergence will serve as the strict validation metric, guaranteeing that the

extracted color centroids accurately and optimally represent the true perceptual variance of the master design template.

To prevent the typographic hallucinations inherent in traditional Optical Character Recognition (OCR) systems, the correctness of the user-defined typographic parameters (font family, font weight, and baseline geometric sizing) will be strictly enforced through pre-orchestration JSON Schema Validation. Prior to initializing the Graph Neural Network (GNN) or the Cassowary solver, the framework will parse the ingested Dynamic Content Payload against a rigid, predefined data schema. The validation algorithm will verify that all typographic inputs strictly conform to expected programmatic constraints (e.g., ensuring font_weight is evaluated as a valid integer and baseline_tensor arrays do not exceed the viewport parameters). If a user payload violates these structural boundaries, the system will programmatically reject the initialization, thereby guaranteeing deterministic typographic correctness and preventing downstream layout failure.

Quantitative Layout Evaluation (Heuristic Fitness Function)

To programmatically evaluate the structural and aesthetic validity of the generated coordinate layouts without relying on subjective human surveying, the framework implements a mathematically rigorous Heuristic Fitness Function. This objective function is formulated as a minimization problem, calculating an aggregate penalty score based on four distinct spatial constraints. The overarching fitness function is defined in Equation 3:

$$F_{layout} = \alpha P_{collision} + \beta P_{alignment} + \gamma P_{whitespace} + \delta P_{legibility} \quad (3.11)$$

In Formula 11, the coefficients $(\alpha, \beta, \gamma, \delta)$ represent static scalar weights empirically calibrated during Phase 3 testing to prioritize critical layout rules (e.g., α is assigned the

highest weight to strictly penalize element overlap). The individual penalty variables are mathematically defined as follows:

1. **Collision Penalty ($P_{collision}$):** This metric calculates the absolute area of spatial overlap between all independently generated bounding boxes (B). It ensures elements do not visually occlude one another, defined in Formula 12:

$$P_{collision} = \sum_{i=1}^{N} \sum_{j=i+1}^{N} Area(B_i \cap B_j) \tag{3.12}$$

2. **Alignment Penalty ($P_{alignment}$):** This metric calculates the spatial variance of generated nodes against a dynamically generated structural grid (G). It computes the minimum pixel distance from a node's baseline coordinate (x_i or y_i) to the nearest gridline, defined in Formula 13:

$$P_{alignment} = \sum_{i=1}^{N} \min_{g \in G} |coordinate_i - g| \tag{3.13}$$

3. **Whitespace Penalty ($P_{whitespace}$):** This metric evaluates the mathematical ratio of negative space to positive design elements, measuring its deviation from an optimal target ratio (R_{target}) established by the master template's visual DNA. It is defined in Formula 14:

$$P_{whitespace} = \left| \frac{Area_{canvas} - \sum_{i=1}^{N} Area(B_i)}{Area_{canvas}} - R_{target} \right| \tag{3.14}$$

4. **Legibility Penalty ($P_{legibility}$):** This metric prevents text degradation by cross-referencing the predicted bounding boxes of text nodes (B_t) against the background Saliency Map (S) generated via SIFT/SURF. It calculates the sum of high-frequency visual noise pixels (p) underlying the text coordinates, defined in Formula 15:

$$P_{legibility} = \sum_{t \in TextNodes} \sum_{p \in B_t} S(p) \tag{3.15}$$

By calculating these interconnected formulas, the orchestration engine mathematically discards invalid spatial configurations and exclusively presents layout candidates that approach the optimal F_{layout} minimum.

Materials and Evaluation Methods

This section will outline the comprehensive methodology utilized to construct, validate, and empirically evaluate the ThemeSync orchestration framework. It will detail the specific computational hardware, software environments, and foundational datasets required as the primary research instruments. Furthermore, this section will define the systematic procedures for both the automated algorithmic testing and the human-in-the-loop usability validation, alongside the targeted sampling techniques applied to the computational data. Finally, it will establish the precise statistical tools and overarching evaluation methods necessary to quantitatively measure algorithmic latency, constraint satisfaction rates, and user interaction efficacy, ensuring a rigorous and multi-dimensional assessment of the proposed architecture

Research Instruments and Data Gathering

The primary instruments for this algorithmic study will comprise specialized computational datasets and structured vector templates. These instruments will provide the foundational data required to train the relational logic models and empirically test the real-time constraint solvers.

Training Data and Feature Mapping

To process visual layouts computationally and establish implicit spatial knowledge, the framework will utilize the CGL-Dataset V2, hosted on the Hugging Face repository, as its primary offline training corpus. This dataset contains thousands of annotated graphic

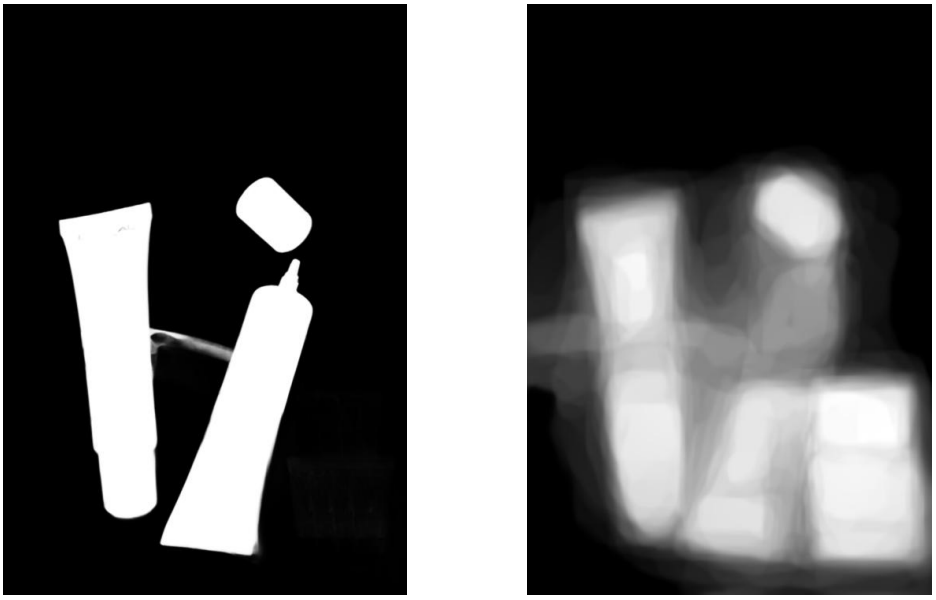
design layouts, providing the necessary variety of structural bounding boxes and semantic categorizations required to train the Graph Neural Network (GNN).

Because of the raw CGL-Dataset is structured in a tabular delivery format containing comprehensive administrative metadata (e.g., image_id, file_name, and nested annotations), the framework will enforce a deterministic preprocessing pipeline prior to neural network ingestion. This preprocessing module will systematically execute three distinct operations. First, the system will perform noise reduction by programmatically discarding non-mathematical administrative columns to optimize tensor processing. Following this, the module will execute JSON deserialization to unpack nested arrays within the annotations and text_annotations columns, thereby extracting precise bounding box geometries (bbox) and baseline semantic identities (supercategory). Finally, the system will perform saliency mapping by cross-referencing the extracted coordinate boundaries against the background saliency matrix (as illustrated in Figure 1). This operation calculates a discrete numerical penalty weight, replacing the need to feed computationally heavy, raw spatial matrices directly into the GNN..

The culmination of this preprocessing pipeline will yield a structurally filtered node array, formatted precisely to populate the GNN's initial feature tensor ($h^{(0)}$). A representative sample of this structured training dataset, demonstrating both structural boundaries and extracted algorithmic features, is detailed in Annex Appendix A.



(a) Original annotated layout (b) Inpainted background isolating the canvas



(c) Saliency map output (d) Saliency map utilized for preprocessing

Figure 1: **CGL-Dataset Visual Extraction Pipeline.**

(a) The original annotated layout. (b) The inpainted background isolating the canvas. (c & d) The corresponding saliency maps utilized by the preprocessing module to calculate the $P_{legibility}$ visual noise penalty weight prior to GNN ingestion.

To guarantee deterministic execution and mathematical rigor, these preprocessed offline dataset arrays, alongside the runtime administrative typographic constraints defined in the Hybrid Visual DNA, must be explicitly mapped to the mathematical variables established within the system architecture. Table 1 demonstrates this precise correlation between the data structures and the algorithmic pipeline.

While the offline dataset establishes the baseline spatial intelligence, the framework will also require a secondary, dynamic data stream during runtime orchestration. Because the framework is specifically designed to handle distinct administrative domains—such as Identification Cards and Academic Certificates—the system must ingest the user's structural requirements prior to final coordinate calculation.

Rather than injecting hardcoded data records at this stage, the system will ingest a structural schema. This lightweight JSON payload will define the expected structural classes of the variables (e.g., text nodes or repeating geometric groups). Crucially, to align with the GNN's implicit geometric hierarchy, the schema will explicitly assign a Spatial Hierarchy Index to each variable. This mathematical weight instructs the Cassowary constraint solver to allocate the largest baseline geometric tensors to primary nodes (e.g., ensuring a student's name receives a larger bounding box than a course subtitle). This mechanism allows the system to assign worst-case, hierarchically accurate geometric tensors to the placeholders, ensuring the layout remains unbroken during future data population. A sample representation of this runtime schema payload, utilizing an academic certificate context, is structured and shown in Annex Appendix B:

To guarantee deterministic execution and mathematical rigor, both the preprocessed dataset arrays and the dynamic payload structures must be explicitly mapped to the algorithmic variables established within the system architecture. Table 1 demonstrates this precise correlation.

Table 1
Data-to-Algorithm Parameter Mapping

Ingested Data Field	Algorithmic Application	Mathematical Variable	Processing Module
`bbox [x, y]`	Top-left spatial coordinate	(x_i, y_i)	GNN Initial Feature Encoding
`bbox [w, h]`	Baseline geometric tensor	(w_i, h_i)	Implicit Geometric Hierarchy
`extracted_color_centroids`	Algorithmic palette mapping	c_i	Node Feature Aggregation
`administrative_typography (font_family, weight)`	Strict brand typographic compliance	S_i	Pre-Orchestration Schema Validation
`extracted_saliency_weight`	Visual noise penalty metric	$p_i(\text{legibility})$	Heuristic Optimization Loop
`spatial_hierarchy`	Tensor allocation priority	Placeholder Tensor	Dynamic Node Instantiation

Synthetic Administrative Templates and Vector Assets

To rigorously evaluate the framework's real-time orchestration engine, a controlled dataset of synthetic administrative templates will be engineered. Rather than utilizing abstract graphic structures, these master files will be explicitly modeled to represent high-constraint institutional domains, specifically Academic Certificates and Student Identification Cards. These templates will be programmatically structured as interconnected Scalable Vector Graphics (SVG) and JSON schemas, designed to encompass varying degrees of spatial complexity.

Crucially, these synthetic assets will act as the primary testing instruments to stress-test the Cassowary constraint solver. To simulate real-world Human-in-the-Loop (HITL) injection, the synthetic dataset will generate dynamic payloads containing extreme spatial variables—such as unusually long string values for student names or maximum instantiation limits for repeating_groups (e.g., signature blocks). This controlled injection will validate the solver's capacity to execute dynamic aspect ratio transformations, correctly enforce

the mathematical spatial_hierarchy weights, and dynamically resize text elements without violating predefined bounding box limits or overwriting surrounding structural nodes.

Research Procedure

The algorithmic pipeline will be executed in three distinct developmental phases, mapping directly to the proposed tri-layer orchestration architecture. This procedural approach will ensure the systematic integration of the perceptual extraction models, the real-time constraint solvers, and the quantitative evaluation metrics.

Phase 1: Data Extraction and Graph Construction

In the initial phase, the Deterministic Style Extraction modules will be programmed to autonomously parse master design templates. The system will utilize K-Means Clustering and SIFT/SURF algorithms to mathematically isolate dominant color centroids and scale-invariant stylistic markers. Concurrently, the system will execute Semantic Layout Parsing utilizing YOLOv8 for object bounding box detection and Tesseract OCR for text extraction, converting the flattened raster pixels into structured, editable nodes. Concurrently, the Graph Neural Network (GNN) architecture will be constructed and trained using the CGL-Dataset. The GNN will be trained to map these extracted visual elements into a formal mathematical graph, establishing the hierarchical node-edge dependencies and spatial reasoning baselines required for the layout generation.

This interface (Figure 2) will demonstrate the initial data extraction and administrative configuration phase prior to layout synthesis. The central viewport will display the uploaded master template, while the right-hand panel will expose the system's underlying mathematical diagnostics and input controls. This unified diagnostic panel will explicitly present the "Hybrid Visual DNA," logically partitioned into two components: the autonomous algorithmic extraction (including dominant color centroids identified via K-Means clustering, boolean feature detection via SIFT/SURF, and the structural hierarchy

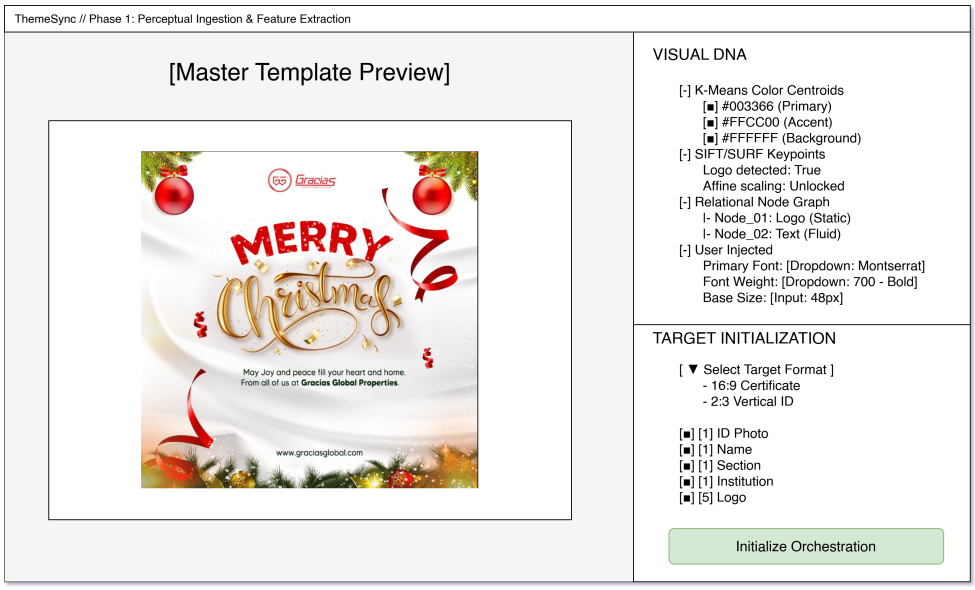


Figure 2: **Perceptual Ingestion and Initialization Dashboard.**

parsed by the GNN) and the administrative typography (where the user deterministically injects font families, font weights, and baseline geometric sizing). By allowing the designer to review the machine-extracted variables, define their strict typographic constraints, and select the target mathematical aspect ratio on a single workspace, the framework will guarantee that all necessary constraints and spatial boundaries are structurally established before initializing the orchestration engine.

Phase 2: Constraint Satisfaction Problem (CSP) Integration and State Synchronization

Following the construction of the relational graph, the second phase will focus on real-time reactive layout synthesis. Prior to spatial calculation, the framework will execute Dynamic Node Instantiation. The system will ingest the user's Content Payload (external administrative datasets formatted as JSON) and algorithmically assign Baseline Geometric Tensors to the new variables, establishing their Implicit Geometric Hierarchy within the graph. The Cassowary Algorithm will then be mathematically implemented within the orchestration layer to act as the primary linear arithmetic constraint solver. Constraint

equations will be formulated and embedded to ensure bounding boxes dynamically adapt to varying aspect ratios without spatial overlap. Simultaneously, the Bi-directional State Machine will be coded to serialize ongoing project states into lightweight JSON formats. This synchronization layer will be explicitly programmed to capture manual user interventions from the Human-in-the-Loop (HITL) interface, translating them into hardened mathematical constraints to permanently prevent destructive overwriting during automated global updates.

Phase 3: Heuristic Testing and Optimization

The final procedural phase will involve the rigorous testing and refinement of the generated coordinate layouts. The system will be subjected to the synthetic administrative templates to execute parallel state-space explorations, effectively generating multiple layout candidates. These algorithmic outputs will be mathematically evaluated against the established heuristic Fitness Function. The system's penalty weights—specifically the collision, alignment, whitespace, and Legibility penalties (derived from cross-referencing text coordinates against the SIFT/SURF Saliency Map)—will be iteratively calibrated. This optimization loop will continue until the mathematical scores demonstrate that the generated coordinate predictions consistently align with strict architectural constraints and human-centric design rules. Once the spatial coordinates are heuristically validated, the Aesthetic Rendering Pipeline will execute. This final algorithmic step will project the K-Means color clusters and extracted stylistic typography onto the generated bounding boxes, producing a fully populated, visually harmonized graphic layout.



Figure 3: **Active Orchestration and Heuristic Scoring Workspace**

This unified workspace (Figure 3) will illustrate the real-time execution of the Cassowary constraint solver, the schema validation protocols, and the aesthetic evaluation algorithms. The central canvas will display the design adapting to the new target format. The right-hand interface will be segmented into two critical computational tabs. The ”Constraint Editor” tab will expose the solver’s active parameters, displaying both the mathe-

matical spatial coordinates (X, Y, W, H) and the schema-validated typographic parameters for the selected node. This allows the designer to dictate whether specific elements are processed as algorithmic ”Soft” constraints or mathematically locked ”Hard” constraints. Below this, a serialized JSON timeline will visually represent the Bi-directional State Machine, explicitly logging the successful Schema Validation of the user's typography and proving the non-destructive retention of manual user refinements. Concurrently, the ”Recommendation Engine” tab will display parallel algorithmic layout variations, explicitly ranked using the Heuristic Fitness Function to validate outputs against objective collision, alignment, and whitespace penalty scores.

Phase 4: Expert Aesthetic Validation

Following the algorithmic generation and mathematical validation of the spatial coordinates in Phase 3, the orchestration framework will undergo a final human-in-the-loop expert evaluation to ensure industry-standard aesthetic compliance. A designated domain expert—defined as a professional layout artist or graphic designer with documented industry experience—will interact with the ThemeSync outputs. The expert will be tasked with reviewing a randomized subset of the algorithmically generated multi-format layouts, specifically the synthetic Academic Certificates and Student Identification Cards, and comparing them against the original Master Layout.

Rather than modifying the underlying computational architecture, the expert will strictly evaluate the final visual outputs. They will utilize a standardized Mean Opinion Score (MOS) rubric to quantitatively grade each generated layout. This evaluation will measure critical design parameters that extend beyond absolute mathematical bounds, effectively bridging the gap between pure computational constraint satisfaction and subjective human design validity. By framing the layout artist's evaluation within a structured numerical rubric, the study will maintain its rigorous quantitative methodology while fulfilling the necessity for real-world professional validation.

Evaluation Metrics

The validation of the orchestration framework will utilize a mixed-methods approach. The core system architecture will rely on controlled, quantitative metrics to measure algorithmic precision, spatial prediction accuracy, and computational efficiency. However, to ensure that the mathematically optimized layouts translate into viable professional designs, this rigorous computational benchmarking will be followed by a structured, quantifiable evaluation by a human domain expert. This dual-validation methodology ensures that the framework achieves both mathematical constraint satisfaction and real-world aesthetic validity.

Intersection over Union (IoU)

Intersection over Union (IoU) will be strictly confined to Phase 1 of the methodology to evaluate the extraction accuracy of the Semantic Layout Parsing module. During the initial ingestion of the Master Template, this metric will calculate the area of overlap between the predicted bounding boxes generated by the YOLOv8 object detection model and the ground-truth annotations established in the baseline dataset. By dividing the area of overlap by the area of union, the system will yield a strict, quantifiable score defining the exact geometric fidelity of the extracted nodes prior to relational mapping. Crucially, IoU will not be utilized to evaluate the final dynamically generated layouts in Phase 3 or Phase 4, as orchestrating nodes across heterogeneous target aspect ratios inherently precludes the existence of a static ground-truth coordinate map.

Algorithmic Latency

This metric will measure the time complexity and execution speed of the underlying algorithms, particularly the Cassowary Algorithm and the Bi-directional State Machine. Algorithmic latency will be measured in milliseconds to mathematically demonstrate that

constraint recalculations and state synchronizations execute within the strict time thresholds required for real-time Human-in-the-Loop (HITL) interface responsiveness, proving that the computational overhead does not bottleneck the orchestration loop.

Structural Accuracy Rate (F1-Score and Classification Metrics)

The overall reliability of the reactive layout synthesis engine (Phases 2 and 3) will be scored utilizing an F1-Score, derived from the harmonic mean of precision and recall. Because generating layouts for heterogeneous aspect ratios precludes the existence of a ground truth, the classification parameters for this metric will be strictly mathematically defined by Constraint Satisfaction rather than spatial overlap.

A True Positive (TP) will be recorded when the algorithm successfully generates and positions bounding box coordinates for a required semantic node, provided that the node mathematically satisfies all designated constraints (i.e., it triggers zero collision penalties and strictly respects user-defined Hard Constraints).

Conversely, a False Positive (FP) will be registered if the algorithm synthesizes bounding box coordinates that violate a mathematically defined spatial boundary. This includes invoking a collision penalty by overlapping an immutable designer-locked node, or protruding beyond the absolute dimensions of the target viewport.

Finally, a False Negative (FN) will occur when the algorithm completely fails to generate bounding box coordinates for a semantic node that is mathematically mandated by the Dynamic Content Payload or the relational graph of the master template, resulting in a missing structural element on the final design canvas.

The Heuristic Fitness Function

Layout evaluation will be framed strictly as a mathematical optimization problem aiming to minimize the aggregate penalty score defined in Formula 11. The heuristic fitness function will objectively score parallel layout candidates by computing the Collision

Penalty (Formula 12 to calculate overlapping nodal areas), the Alignment Penalty (Formula 13 to measure pixel variance from established gridlines), the Whitespace Penalty (Formula 14 to evaluate the mathematical ratio of negative space), and the Legibility Penalty (Formula 15 to quantify visual interference by cross-referencing text coordinates against the high-frequency keypoints of the SIFT/SURF Saliency Map). By utilizing these explicit mathematical thresholds, the framework guarantees that subjective aesthetic principles are translated into rigorous, machine-readable metrics, thereby eliminating the necessity for subjective human surveying to validate structural accuracy.

Mean Opinion Score (MOS) for Aesthetic Alignment

To execute the expert aesthetic validation in Phase 4, the designated layout expert will evaluate the generated multi-format layouts using a structured 5-point Likert scale. This Mean Opinion Score (MOS) will quantify the subjective aesthetic validity of the final designs into rigorous, measurable data. The expert will evaluate a randomized sample of the generated layouts against three core design parameters:

1. **Visual Hierarchy Preservation:** Evaluates whether the algorithmic coordinate placement maintains the correct structural dominance (e.g., ensuring a student's name or a certificate title visually remains the primary focal point relative to the surrounding nodes).

2. **Brand and Typographic Consistency:** Evaluates the framework's success in enforcing the Hybrid Visual DNA, specifically grading the accurate rendering of user-defined fonts, baseline geometric sizes, and algorithmically extracted color palettes without hallucination.

3. **Constraint and Whitespace Harmony:** Evaluates the subjective quality of the spatial arrangement, confirming the visual absence of element occlusion (overlapping

text/graphics) and verifying that the layout's negative space aligns with professional composition standards.

For each parameter, the expert will assign a numerical score based on the MOS rating scale shown in Table 2. The aggregate MOS across the tested dataset will serve as the final

Table 2
Mean Opinion Score Rating Scale for Aesthetic Alignment

Score	Rating	Description
5	Excellent	No noticeable flaws
4	Good	Minor aesthetic deviations
3	Acceptable	Noticeable but non-destructive deviations
2	Poor	Significant layout fragility
1	Fail	Severe structural overlap or typographic failure

validation metric, mathematically confirming that the system's heuristic constraint solvers successfully translate structural boundaries into industry-viable graphic assets.

Theoretical and Conceptual Framework

This section will establish the foundational theoretical principles and the overarching conceptual paradigm that govern the architecture and evaluation of the ThemeSync orchestration framework. It will delineate the core computational theories underpinning the system, specifically drawing upon constraint-based layout theory to justify the mathematical resolution of spatial boundaries, alongside computational aesthetic theories to ground the machine-learning-driven evaluation models. Furthermore, this section will present the conceptual framework, explicitly mapping the systematic interaction of variables throughout the execution pipeline. It will illustrate the continuous data flow from the initial ingestion of unstructured graphical layouts and user-defined parameters, through the concurrent processing of the neural network and mathematical modules, to the final output of stabilized, compositionally aligned design candidates. By synthesizing these theoretical and

conceptual constructs, this section provides the rigorous academic scaffolding required to mathematically and logically justify the system's structural design and subsequent empirical methodology.

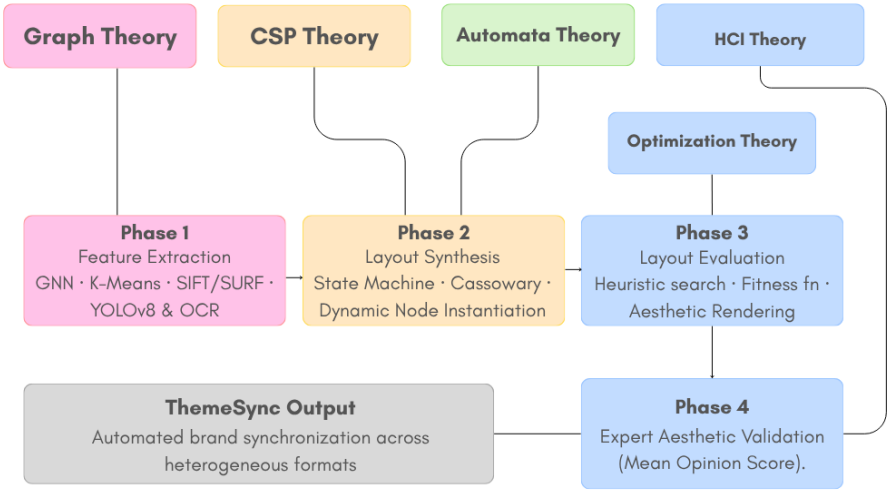


Figure 4: **Theoretical Framework Diagram**

Figure 4 illustrates the theoritical framework that underpins this research. The ThemeSync framework is grounded in four fundamental computer science theories: Graph Theory, the Constraint Satisfaction Problem (CSP) paradigm, Automata Theory, and Optimization Theory. These interconnected theoretical models provide the formal mathematical logic required to transition graphic layout orchestration from a static, coordinate-based generation task into a dynamic, stateful algorithmic pipeline.

Graph Theory

Graph Theory serves as the primary theoretical basis for Phase 1 of the system, governing the relational logic of the layout components. In this context, a graphic layout is not modeled as a flattened array of pixels, but rather as a highly structured mathematical graph. Individual design elements, such as text blocks, image containers, and background shapes, operate as nodes, while the spatial and structural dependencies between them—such as

proximity, alignment, and hierarchical nesting—operate as edges. This theory justifies the application of Graph Neural Networks (GNN) and configurable graph-based frameworks within the system, establishing that deep learning models can accurately parse, traverse, and reconstruct the topological anatomy of a design by mathematically evaluating these node-edge relationships prior to rendering [16]. Furthermore, integrating perceptual extraction techniques, such as algorithmic palette mapping, directly into this graph ensures that stylistic and color identities remain mathematically consistent during structural transformations [7].

Constraint Satisfaction Problem (CSP) Paradigm

The CSP paradigm underpins Phase 2 by providing the theoretical basis for reactive layout synthesis. CSP defines problems where the state of a set of objects must satisfy a specific number of logical or arithmetic constraints. Applied to spatial orchestration, responsive layout generation transforms into a complex equation-solving environment capable of supporting real-time GUI interactions [11]. Viewport dimensions and target aspect ratios are treated as mathematical variables, while structural rules and user intents are encoded as linear inequalities and adaptive OR-constraints [2]. This foundation governs the implementation of scalable constraint generation models [4, 12], proving that dynamic spatial harmony can be mathematically enforced in real time — eliminating the need for stochastic pixel placement.

Automata Theory

To address the persistent issue of destructive layout generation, the framework integrates principles from Automata Theory and state-machine logic. Automata Theory is the study of abstract machines and systems that transition between distinct computational states based on defined inputs. Within ThemeSync, this grounds the Bi-directional State Machine and its corresponding JSON schema handling. It provides the formal logical

proof that the orchestration engine can maintain a persistent ”memory” of localized manual edits—treating a user’s manual refinement as a highly specific computational state—and seamlessly transition the global template structure to a new iteration without irreversibly overwriting the human-in-the-loop modifications.

Optimization Theory

Optimization Theory supports Phase 3 of the system, governing the algorithmic evaluation of the generated layouts. Optimization Theory focuses on finding the best available mathematical solution from a massive set of feasible, parallel alternatives. In the context of automated orchestration, determining the optimal layout requires navigating a rigorous mathematical search space. This theory justifies the system’s Heuristic Search mechanism, establishing that subjective aesthetic composition rules—such as geometric balance, whitespace distribution, and margin collision avoidance—can be quantified into objective penalty scores. By minimizing these penalty scores via a mathematical Fitness Function, the system successfully bridges the gap between automated spatial calculation and human-centric aesthetic validity.

Human-Computer Interaction (HCI) Theory

Finally, to support the Phase 4, the framework draws upon Human-Computer Interaction (HCI) theory, which asserts that computational systems must align with human cognitive and perceptual standards to achieve practical utility. While Optimization Theory governs the mathematical minimization of spatial errors, HCI theory justifies the necessity of the Mean Opinion Score (MOS). It provides the theoretical grounding that pure mathematical constraint satisfaction is insufficient for creative orchestration without human-in-the-loop validation. By structuring the expert’s subjective aesthetic evaluation into a quantifiable metric, the system proves that algorithmic layout generation can successfully synthesize mathematical precision with human-centric design expectations.

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APPENDICES

APPENDIX A

PREPROCESSED CGL-DATASET NODE ARRAY FOR GNN INGESTION


```
[
  {
    "node_id": "19",
    "semantic_category": "logo",
    "bbox": [106, 22, 284, 65],
    "extracted_color_centroids": ["#E83F3F", "#FFFFFF"],
    "extracted_saliency_weight": 0.45
  },
  {
    "node_id": "20",
    "semantic_category": "text",
    "bbox": [61, 140, 393, 67],
    "text_features": {
      "content": "11.1\\u62a2\\u534a\\u4ef7",
      "font_size_scale": 0.08,
      "alignment": "left"
    },
    "extracted_saliency_weight": 0.72
  }
]
```

APPENDIX B

DYNAMIC CONTENT PAYLOAD FOR REAL-TIME RUNTIME ORCHESTRATION

```
{
  "domain_context": "academic_certificate",
  "target_aspect_ratio": "4:3",
  "variable_schema": {
    "{{Student_Name}}": {
        "structural_class": "text_node",
        "spatial_hierarchy": 1,
        "max_characters": 60
    },
    "{{Course_Title}}": {
        "structural_class": "text_node",
        "spatial_hierarchy": 2,
        "max_characters": 80
    },
    "{{Signature_Blocks}}": {
        "structural_class": "repeating_group",
        "spatial_hierarchy": 3,
        "instantiation_count": 3
    }
  }
}
```